Three involutions in finite general linear groups

A complete principal-result recovery of Markovskaya and Nuzhin (2025)

We recover the two classifications of I. A. Markovskaya and Ya. N. Nuzhin [MN], retaining their explicitly borrowed classifications and integral generation theorem. Their constructive matrices are retained and their root-generation steps are supplied by the complete common paper *Root parameters and fixed-space obstructions* [U]. We correct three displayed matrix expressions below. These corrections do not contradict either principal classification. The intended domain is $n \geq 2$, as in both source tables; the literal omission of that domain in the theorem sentences is recorded separately.

1 Definitions and retained inputs

Call a group admissible if it is generated by three elements of order at most two, the first two commuting. For a nontrivial group this is equivalent to the source convention allowing repeated involutions: an identity entry can be replaced by an existing suitable involution (if a commuting entry is identity, repeat the other commuting entry; if both are identities, repeat the remaining entry). The trivial group is included by convention. Thus all dihedral groups are admissible. Quotients are admissible: simply take the images of the marked generators. For $H \lhd G$, a tuple generates G/H exactly when it generates G together with H, by taking the full inverse image of the generated quotient subgroup.

The following inputs are retained, not reclassified in this paper.

1. The known classification of admissible finite simple linear groups, as used in [MN, introduction and proof of Theorem 2]: among $\mathrm{PSL}_n(q)$, $n \geq 2$, the exclusions are $\mathrm{PSL}_2(7)$, $\mathrm{PSL}_2(9)$, every $\mathrm{PSL}_3(q)$, and every $\mathrm{PSL}_4(2^a)$. The small nonsimple groups $\mathrm{PSL}_2(2)$ and $\mathrm{PSL}_2(3)$ are handled separately if needed. We use this classification only for $n \geq 3$; it is attributed to Nuzhin [N19].

2. $\mathrm{PGL}_2(q)$ is admissible for every prime power $q > 2$, by Sjerve–Cherkassoff [SC]. The remaining $\mathrm{PGL}_2(2) \cong S_3$ is dihedral.

3. $\mathrm{GL}_n(\mathbb{Z})$ is admissible for $n \geq 5$, by the earlier Markovskaya–Nuzhin theorem [MN23]. Its proof is retained, not replaced by computing finitely many reductions.

4. The rank-three intersection criterion of Conder–Oliveros [CO, Corollary 4.2]: three distinct involutions, two commuting, in a finite group with no nontrivial cyclic normal subgroup give a regular abstract rank-three polytope. The standard group–polytope correspondence is retained with this criterion.

These are genuine dependencies, whose complete external proof and software costs are unknown here. The source's general rank-two transvection statements [MN, Lemma 10], attributed to Levchuk [L83b], are also retained as background statements, but none is a premise of our constructive proof. All root and fixed-space steps actually used below are proved in [U]. For precision the three retained rank-two assertions are: (a) if $q \neq 9$, $V \subseteq \mathbb{F}_q^*$ has at least two elements and contains an element lying in no proper subfield, then $\langle t_{12}(V), t_{21}(1)\rangle = \mathrm{SL}_2(q)$; (b) for nonzero a, b, the group $\langle t_{12}(a), t_{21}(b)\rangle$ contains some $\left(\begin{smallmatrix} 0 & -w^{-1} \\ w & 0 \end{smallmatrix}\right)$ and its square $-I$; (c) for odd q and primitive u, $\langle t_{12}(u), t_{21}(1)\rangle = \mathrm{SL}_2(q)$. These retain the source's exact hypotheses and are not claimed as consequences of our recognition theorem, which requires at least three coordinates.

2 Determinants, quotients, and the linear classification

For a commutative unital ring K, the determinant has the section $a \mapsto \mathrm{diag}(a, 1, \ldots, 1)$, and hence

$$\mathrm{GL}_n(K) = \mathrm{SL}_n(K) \rtimes \{\mathrm{diag}(a, 1, \ldots, 1) : a \in K^*\}.$$

This argument does not require elementary generation over K. If K^* is finite and $\gcd(n, |K^*|) = 1$, its nth-power map is an automorphism. Every determinant is therefore the determinant of a scalar matrix, with trivial intersection between scalars and $\mathrm{SL}_n(K)$. Thus $\mathrm{GL}_n(K) = \mathrm{SL}_n(K) \times K^* I$ and $\mathrm{PGL}_n(K) \cong \mathrm{SL}_n(K) = \mathrm{PSL}_n(K)$.

Over a finite field, Gaussian elimination shows that all elementary transvections generate $\mathrm{SL}_n(q)$. Reduction from $\mathrm{GL}_n(\mathbb{Z})$ modulo a prime p contains all these transvections and $\mathrm{diag}(-1, 1, \ldots, 1)$; its image is exactly the determinant-$\{1, -1\}$ subgroup of $\mathrm{GL}_n(p)$. It is onto precisely for $p = 2, 3$. For $n = 1$ the same determinant description also holds, though that dimension is not in the present classifications.

Theorem 1. *For $n \geq 2$, the group $\mathrm{GL}_n(q)$ is admissible exactly when*

$$(q = 2 \text{ and } (n = 2 \text{ or } n \geq 5)) \quad \text{or} \quad (q = 3 \text{ and } n \geq 5).$$

Proof. For $q \geq 4$, determinants of involutions lie in $\{1, -1\}$, a proper subgroup of $\mathbb{F}_q^*$, so even unrestricted involution generation fails. For $q = 2$, all four linear/projective/special groups coincide. The retained simple classification excludes $n = 3, 4$ and allows $n \geq 5$; $n = 2$ gives S_3. For $q = 3, n \geq 5$, reduce the retained integral generating triple. For $n = 4, q = 3$, [U, symmetric-square obstruction] supplies the full ten-dimensional argument. For $n = 3, q = 3$, the direct product just proved has $\mathrm{SL}_3(3) = \mathrm{PSL}_3(3)$ as a quotient, which is excluded by the retained classification.

It remains to exclude $\mathrm{GL}_2(3)$. Commuting involutions are simultaneously diagonalizable, so place the commuting pair in the diagonal group. Every nonscalar involution is $\left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right)$ with $a^2 + bc = 1$. If $a = \pm 1$, then $bc = 0$, so it is upper or lower triangular; if $a = 0$, then $bc = 1$, so it is monomial. A diagonal group together with one such matrix remains in the corresponding triangular or monomial proper subgroup. Scalar possibilities cannot enlarge this subgroup. □

The determinant quotient of $\mathrm{PGL}_n(q)$ is

$$\mathrm{PGL}_n(q)/\mathrm{PSL}_n(q) \cong \mathbb{F}_q^*/(\mathbb{F}_q^*)^n \cong C_{\gcd(n, q-1)}.$$

An abelian group generated by involutions has exponent at most two. Hence if $\gcd(n, q-1) > 2$, $\mathrm{PGL}_n(q)$ is not generated by any collection of involutions. This proves [MN, Lemma 11], including its stated divisible special case, without restrictions on lifts of projective involutions.

3 Common coordinates for even dimensions

For the remaining case put $n = 2k \geq 4$, q odd, and $\gcd(n, q-1) = 2$. Choose a primitive $u \in \mathbb{F}_q^*$ and set $\epsilon = (-1)^k$. All matrices act on column vectors. Let

$$\mu e_i = e_{i+1} \quad (i \bmod n), \qquad \tau e_i = e_{n+1-i},$$
$$B = \mathrm{diag}(\underbrace{\epsilon/u, \ldots, \epsilon/u}_{k}, \underbrace{1, \ldots, 1}_{k})\tau, \qquad C = \tau\mu, \quad E = BC.$$

Then $B^2 = (\epsilon/u)I$, $C^2 = I$, and E is the displayed diagonal matrix times μ. With cyclic indices and its diagonal entries d_i,

$$t_{ij}(a)^E = t_{i+1,j+1}(a\, d_{i+1}/d_{j+1}). \tag{1}$$

The conjugation convention throughout is $X^Y = YXY^{-1}$. Directly,

$$\det B = u^{-k}, \qquad \det C = (-1)^{k-1}.$$

If k is odd, $\gcd(k, q-1) = 1$, so $\det B$ is primitive. If k is even, write $q-1 = 2m$ and $k = 2h$. The gcd assumption gives m odd and $\gcd(h, m) = 1$; $\det(BC) = -u^{-k} = u^{m-2h}$ is primitive, since $\gcd(m-2h, 2m) = 1$. Thus once $\mathrm{SL}_n(q)$ lies in $\langle A, B, C\rangle$, this group is the whole $\mathrm{GL}_n(q)$ in the cases using these coordinates.

3.1 All dimensions at least eight

Let A have diagonal entries -1 at $2, 3, n-2, n-1$, all other diagonal entries 1, and just two off-diagonal entries $A_{21} = A_{n-1,n} = 1$. Then $A^2 = I$ and $AB = BA$. Define the following words without changing their order:

$$X = [A, A^E], \quad T = ((XA^{E^2})^2)^E, \quad Z = [T, X], \quad R = [A, Z], \quad V = R^B, \quad W = [Z, V].$$

The complete nonidentity entries of these words are as follows; products in each table entry have commuting factors.

word	value
X	$t_{31}(-1)t_{n-1,1}(\epsilon u)$
T, $n = 8$	$t_{43}(-u)t_{52}(u)t_{53}(-u^2)t_{83}(u^3)$
T, $n \geq 10$	$t_{43}(-\epsilon u)t_{52}(1)t_{53}(-\epsilon u)t_{n3}(\epsilon u^3)$
Z, $n = 8$	$t_{41}(u)t_{51}(u^2)t_{81}(-u^3)$
Z, $n \geq 10$	$t_{41}(\epsilon u)t_{51}(\epsilon u)t_{n1}(-\epsilon u^3)$
R	$t_{n-1,1}(-\epsilon u^3)$
V	$t_{2n}(-u^2)$
W	$t_{21}(-\epsilon u^5)$

These identities hold as Laurent-polynomial identities over $\mathbb{Z}[u, u^{-1}]$, with $\epsilon = (-1)^k$. One way to verify them uniformly is to conjugate the diagonal entries and the two off-diagonal entries of A using (1), and multiply using $E_{ij}E_{ab} = \delta_{ja}E_{ib}$. Only indices $1, 2, 3, 4, 5, n-2, n-1, n$ occur. The distinction in the two T, Z rows is exactly that $d_5 = 1$ for $n = 8$, whereas $d_5 = \epsilon/u$ for $n \geq 10$. No enumeration of dimensions or fields is a premise of these identities.

The nonzero adjacent root W and the monomial cycle E reach every root by [U, root propagation]. Apply [U, finite root-parameter recognition] to $M = \langle A, B, C\rangle$. Its root-generated subgroup contains the determinant-one diagonal sign matrices. Isolating the second factor of X by [U, isolation] gives $t_{n-1,1}(\epsilon u)$. Together with R this puts $-u^2$ in the common parameter subfield; thus that subfield is the whole field, by [U, primitive-square test]. Hence M contains $\mathrm{SL}_n(q)$ and is $\mathrm{GL}_n(q)$ by the determinant argument. The projective images of A, B, C are involutions, the first two commute, and they generate $\mathrm{PGL}_n(q)$.

3.2 Dimension six

For $q = 3$ Theorem 1 already supplies the required projective triple. Otherwise choose the primitive element so that $h = u^2 + 4u \neq 0$. This is possible: the only forbidden nonzero value is -4, and a cyclic group of even order $q - 1 \geq 4$ has at least two primitive elements. Keep B, C, E above, and let A have diagonal entries -1 at $1, 6$, 1 elsewhere, and $A_{21} = A_{56} = -1$. Again $A^2 = I$ and $AB = BA$. The exact words are

$$\begin{aligned} X = [A, A^E] &= t_{31}(-1)t_{51}(-u), \\ Q = [A^{E^4}, X] &= t_{51}(2u)t_{61}(-u^2), \\ [A, Q] &= t_{51}(-h), \\ (A^{E^4}t_{51}(h))^2 &= t_{61}(uh), \qquad t_{61}(uh)^E = t_{12}(uh). \end{aligned}$$

These follow from (1) and the same matrix-unit product rule. Since $uh \neq 0$, propagation again gives all roots. Root recognition then supplies $D = \mathrm{diag}(1, 1, 1, 1, -1, -1)$, and

$$[D, X] = t_{51}(2u), \qquad t_{51}(2u)^C = t_{15}(2u).$$

Their opposite-root parameter product is $4u^2$; the primitive-square test gives the full field. Thus M contains $\mathrm{SL}_6(q)$, and $\det B = u^{-3}$ is primitive because $\gcd(6, q-1) = 2$. This proves the six-dimensional branch without assuming that $2u$ itself is a primitive multiplicative element.

4 Dimension four

Here $\gcd(4, q-1) = 2$ means $q \equiv 3 \pmod 4$, so -1 is nonsquare. In both branches put

$$A = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

We will obtain $\mathrm{SL}_4(q) \le M = \langle A, B, G\rangle$, with $\det A = \det B = 1$ and $\det G = -1$. It follows that M is exactly the determinant-$\{1, -1\}$ subgroup. Its projective image is all of $\mathrm{PGL}_4(q)$ because $\mathbb{F}_q^*/(\mathbb{F}_q^*)^4$ has order two and -1 represents its nontrivial coset.

4.1 Characteristic three

Take

$$B = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \qquad G = \begin{pmatrix} -1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -u & 0 & 1 & 0 \\ -u & 0 & 0 & 1 \end{pmatrix}.$$

Then $A^2 = B^2 = -I$, $G^2 = I$, and $AB = -BA$. Thus their projective images are involutions, the first two commuting. Set

$$J = GG^{AB}G, \qquad D = (G^{AB}G)^2 J^B, \qquad T = GDD^A G^A, \qquad P = (T^B J)^2.$$

Multiplication over $\mathbb{F}_3[u]$ gives

$$D = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & u & u \\ 0 & u & -u^2-1 & 1-u^2 \\ -u & 0 & 0 & 1 \end{pmatrix},$$

$$DD^A = \begin{pmatrix} -1 & 0 & 0 & -u \\ -1 & 1 & 0 & -u \\ -u & 0 & 1 & u^2-1 \\ -u & 0 & 0 & -u^2-1 \end{pmatrix},$$

$$T = t_{24}(u)t_{34}(u^2), \qquad P = t_{42}(w), \qquad w = u^3 + u,$$

$$P^{AB} = t_{31}(w), \quad P^A = t_{13}(-w), \quad [G^{AB}, P^{AB}] = t_{32}(w), \quad t_{32}(w)^A = t_{23}(-w).$$

Here $w \neq 0$, because $u^2 = -1$ is impossible. The four displayed roots on coordinates $1, 2, 3$ give a strongly connected directed graph. Applied to the subgroup they generate on these three coordinates, root recognition supplies determinant-one sign matrices and monomial lifts of S_3 on these coordinates, fixing the fourth coordinate. In particular

$$(T\,\mathrm{diag}(-1, 1, -1, 1))^2 = t_{24}(-u), \qquad t_{24}(-u)^B = t_{42}(u).$$

The permutation parts of that S_3 and of AB, namely $(12)(34)$, generate S_4. Hence conjugating a nonzero root reaches all ordered coordinate pairs. Root recognition in dimension four and the opposite parameters u, u now give $\mathrm{SL}_4(q) \le M$.

The additional source assertion about linear lifts is also valid: $\langle uA, B, G\rangle = \mathrm{GL}_4(q)$. Write $q - 1 = 2m$, m odd. Since $(uA)^2 = -u^2 I = u^{m+2} I$ and $\gcd(m + 2, 2m) = 1$, this group contains every scalar, hence A and M. Its determinants generate $\langle -1, u^4\rangle = \mathbb{F}_q^*$.

4.2 Odd characteristic different from three

Take instead

$$B = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad G = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ u & 0 & 1 & 0 \\ u & 0 & 0 & 1 \end{pmatrix}.$$

Here $A^2 = -I$, $B^2 = G^2 = I$, and $AB = -BA$. Let $J = GG^{AB}G$. Its upper-left block is $\left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right)$ and its lower block is I_2. The products JAB and $(JAB)^A$ are the ordinary swaps (34) and (12) respectively. Consequently $D = (JAB)^A J = \mathrm{diag}(-1, -1, 1, 1)$ lies in M, and

$$[D, G] = t_{31}(2u)t_{41}(2u).$$

Taking an integer power with exponent representing $-1/2 \in \mathbb{F}_p$ gives $V = t_{31}(-u)t_{41}(-u)$. Direct multiplication gives

$$E_0 = GV = \mathrm{diag}(-1, 1, 1, 1)t_{21}(1), \qquad U = V^A = t_{14}(u)t_{24}(u), \qquad (UE_0)^2 = t_{24}(3u).$$

As $3 \neq 0$, an integer power gives $t_{24}(u)$ and conjugation by B gives $t_{42}(u)$. The monomial matrices $A, B, (12), (34)$ preserve the partition $\{1, 2\} \mid \{3, 4\}$ and transport this root onto every ordered pair across the partition. This directed complete bipartite graph is strongly connected. Propagation gives all roots, and the opposite parameters give the full field, proving $\mathrm{SL}_4(q) \leq M$. This supplies the source's omitted last generation step without incorrectly assuming that these monomial permutation parts form S_4.

5 Projective classification and both tables

Theorem 2. *For $n \geq 2$, $\mathrm{PGL}_n(q)$ is admissible precisely in these cases:*

$$n = 2; \qquad n \geq 4,\ \gcd(n, q-1) = 2; \qquad n \geq 5,\ \gcd(n, q-1) = 1.$$

Proof. The degree-two case is retained from [SC], with $q = 2$ handled directly. The determinant quotient excludes gcd greater than two. For gcd one, $\mathrm{PGL}_n(q) \cong \mathrm{PSL}_n(q)$, so the retained simple classification gives exactly the asserted possibilities: in degree four gcd one forces q even, and those groups are excluded. For degree three the gcd is either one or three, giving exclusion in both cases. Gcd two forces even $n \geq 4$ and odd q; the preceding three dimension branches of the construction (four, six, and at least eight) exhaust this case. □

group	parameters	admissible
$\mathrm{GL}_n(q)$	$n = 2, q = 2$	yes
	$n = 2, q \neq 2$	no
	$n = 3$, any q	no
	$n = 4$, any q	no
	$n \geq 5, q = 2, 3$	yes
	$n \geq 5, q \neq 2, 3$	no
$\mathrm{PGL}_n(q)$	$n = 2$, any q	yes
	$n = 3$, any q	no
	$n \geq 4, \gcd(n, q-1) > 2$	no
	$n \geq 4, \gcd(n, q-1) = 2$	yes
	$n = 4, \gcd(n, q-1) = 1$	no
	$n \geq 5, \gcd(n, q-1) = 1$	yes

Thus every row of the two source tables follows, with the explicitly retained inputs specified above.

Every positive projective group other than $\mathrm{PGL}_2(2)$ also gives the claimed rank-three regular polytope. The generators can be taken distinct: a repeated pair would make the group dihedral, whereas these groups are not dihedral. For the almost simple cases, a cyclic normal subgroup N would intersect the simple socle trivially and therefore centralize it, since $[N, S] \leq N \cap S$. The centralizer of $\mathrm{PSL}_n(q)$ in $\mathrm{PGL}_n(q)$ is trivial: a projective centralizer fixes each elementary transvection up to a scalar, that scalar must be one by its sole eigenvalue 1, and commuting with all matrix units forces a scalar matrix. The exceptional small positive group $\mathrm{PGL}_2(3) \cong S_4$ has no nontrivial cyclic normal subgroup either, as its normal subgroups are $1, V_4, A_4, S_4$. Apply the retained [CO] criterion. No new answer to the broad problems mentioned in the source's survey citation is claimed.

6 Literal-source qualifications and attribution

1. The source's theorem sentences omit $n \geq 2$, while the two tables begin at degree two. Without that intended domain, the sentences need correction: $\mathrm{PGL}_1(q)$ is trivial for every q, and $\mathrm{GL}_1(2)$ and $\mathrm{GL}_1(3)$ are admissible. No excluded degree-≥ 2 case is changed.

2. On printed p. 254 the intermediate determinant expression for B has a sign inconsistency when k is even. The correct value is u^{-k}, as computed from the full matrix; the primitive determinant argument above covers both parities.

3. On p. 256 the $(4, 4)$ entry of DD^A is $-u^2 - 1$, not $-u - 1$. The subsequent source word for T agrees with the corrected entry.

4. On p. 257, in characteristic different from three, the conjugator in $t_{14}(u)t_{24}(u) = V^A$ must be A, not the printed G: $V^G = t_{31}(u)t_{41}(u)$. The next root identity requires the square of the whole product $(UE_0)^2$; squaring only E_0 leaves U unchanged. These are explicit word corrections, not counterexamples to Theorem 2.

5. The source's root-recognition applications are completed here by forming the root-generated subgroup and isolating paired root factors. In dimension six the opposite-product $4u^2$ argument avoids assuming that $2u$ is multiplicatively primitive.

6. The English end matter prints received/revised dates in 2024, whereas the Russian end matter gives 2025. This is bibliographic only.

All constructive matrices and the symmetric-square obstruction are due to the source authors; our contribution is the complete elementary common proof, explicit parameter extraction and corrected identities. The older private rank-three diagonal-extension work uses different marked groups and is not silently imported here. Symbolic checks of displayed words are optional diagnostics: the written matrix identities, multiplication rule and uniform arguments are the mathematical proof. No new Lean declaration, finite search premise, compression ratio, or conjecture resolution is asserted.

References

[1] [MN] I. A. Markovskaya and Ya. N. Nuzhin, On generation of the groups $\mathrm{GL}_n(q)$ and $\mathrm{PGL}_n(q)$ by three involutions, two of which commute, Trudy Inst. Mat. Mekh. UrO RAN 31(4) (2025), 247–259. DOI 10.21538/0134-4889-2025-31-4-fon-03.

[2] [U] Root parameters and fixed-space obstructions, complete companion manuscript to this recovery.

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